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Direct welding of silicon carbide and fused silica using femtosecond lasers: Effects of repetition rate and focal depth

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Highlights

- Direct welding of silicon carbide and fused silica using femtosecond lasers.
- Element mixing and diffusion under focused laser energy enabled the successful welding of the two materials.
- Island-like structures formed as fused silica adhered to the SiC surface.

Abstract

Outline

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Silicon carbide (SiC) is a wide bandgap semiconductor known for its durability and high performance. These properties make it an ideal material for advanced applications. In this study, a femtosecond laser with a wavelength of 1030nm was employed to weld SiC to fused silica directly. The key experimental variables examined were the laser repetition rate (100, 300, 1000kHz) and focal depth. Energy-dispersive spectroscopy (EDS) was used to analyze the elemental distribution, revealing a spatial gradient in composition. This indicated the mixing and diffusion of elements between the two materials under focused laser energy. Separation specimens showed that fractures occurred primarily within the fused silica near the welded region, where island-like structures formed as fused silica adhered to the SiC surface.

Keywords

Silicon carbide; Fused silica; Femtosecond laser; Multiphoton absorption; Laser welding

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1. Introduction

Silicon carbide (SiC), a third-generation compound semiconductor, is known for its wide bandgap (3.23eV), chemical stability, high breakdown electric field, and exceptional mechanical hardness. These characteristics make it an ideal material for high-performance, high-temperature electronic devices. However, due to SiC's chemical inertness and hardness, conventional microfabrication processes present significant challenges. By heterogeneously welding SiC with other substrate materials, such as silicon (Si) or silicon dioxide (SiO_2), it is possible to not only overcome these fabrication difficulties but also leverage the well-established production technologies of silicon-based materials, which are the most commonly used substrates in electronic circuits.

CRediT authorship contribution statement

Declaration of competing interest

Acknowledgments

Data availability

References

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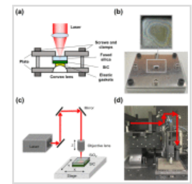
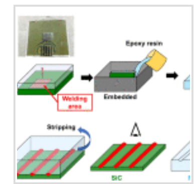
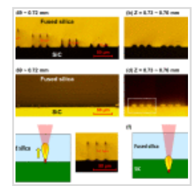
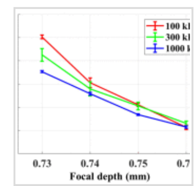
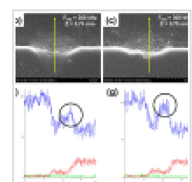
Figures (10)

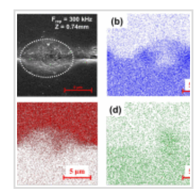












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For instance, heterojunctions created in power devices can lead to improved efficiency. Nonetheless, conventional laser micro-welding, which relies on linear absorption of laser photon energy, presents limitations, especially when welding transparent materials. Typically, an intermediate absorbing layer is inserted between transparent materials to facilitate the absorption of laser energy [1]. However, this layer can induce residual thermal stress due to mismatched thermal expansion coefficients, potentially compromising the weld strength.

Table 3. The effect of parameter adjustm…

Researchers have developed a novel laser micro-welding technique to address these issues based on nonlinear absorption [2]. Nonlinear absorption requires a high laser peak intensity, allowing material melting or ionization to occur only in the focal region. This enables the welding of transparent materials without an intermediate absorbing layer. Tamaki et al. [3], [4] explored femtosecond laser welding of silica glass without an intermediate layer, though a low repetition rate laser (1 kHz) produced limited welding strength. The localized heat accumulation effects could be achieved by increasing the repetition rate to 500 kHz, enabling successful welding between aluminosilicate glass and silicon substrates [4].

Miyamoto et al. [5] further studied the influence of repetition rate (100 kHz to 5 MHz) on crack formation in transparent materials, proposing that by limiting the free surface formation and creating a solid boundary, elasticity in the weld region can suppress shrinkage stress, thereby improving weld quality and preventing cracks [6]. They developed a heat conduction model based on nonlinear absorption through experiments and simulations, explaining the upward expansion of the molten region caused by avalanche ionization, which leads to the formation of a plasma column [7], [8]. In a related study, Huang et al. [9] used a picosecond laser to weld borosilicate glass. Through energy-dispersive spectroscopy (EDS) analysis, they found that the diffusion of silicon (Si) and oxygen (O) elements during welding increased the microhardness of the weld region. This result highlights the essential role of elemental diffusion in enhancing weld quality.

Welding SiC and fused silica is particularly challenging due to their substantial differences in thermal expansion coefficients (SiC: 5.0×10^{-6} K^{-1}; SiO_2: 5.9×10^{-7} K^{-1}) and melting/softening temperatures (SiC: 3100°C; SiO_2: 1720°C) [10]. However, previous studies have demonstrated that managing the free surface can help reduce shrinkage stress, mitigating these difficulties. Zhang et al. [10] employed femtosecond laser multiphoton absorption to weld these two wide bandgap materials (SiC: 3.23 eV; SiO_2: 9.0 eV). They provided initial observations on the morphology and elemental distribution within the weld region, confirming material mixing. However, their study did not explore the effects of parameters such as repetition rate and focal depth on the weld quality.

This study aims to address these gaps by focusing on the femtosecond laser welding of SiC and fused silica, particularly emphasizing the effects of repetition rate and focal depth. Detailed observations of the weld morphology and elemental distribution will provide new insights into the underlying mechanisms governing the welding process.

2. Experimental details and measuring instruments

2.1. Experiment setup

In this study, polished n-type 4H-SiC wafers (UCT4SICDDP010, Atecom Inc.) with a thickness of 350µm were utilized, along with fused silica wafers (KTX Material Inc.) that had a thickness of 1 mm. The welding samples measured 15×15 mm^2 for SiC and 25×25 mm^2 for fused silica. A close contact was established using a fixture to ensure a strong bond between the two materials, as shown in Fig. 1. The fixture applied external pressure at the center of the samples. In Fig. 1(a), a convex lens concentrated the pressure at the center of the sample, while elastic gaskets were used to prevent stress concentrations that could lead to sample breakage. Plates with screws and clamps were used to apply the external pressure required for welding. Before fixing the samples, they were cleaned with alcohol wipes to remove any particles. A close contact was confirmed by the presence of colorful rings in the clamped region. The central area enclosed by these rings represents the primary welding zone, as illustrated in Fig. 1(b).

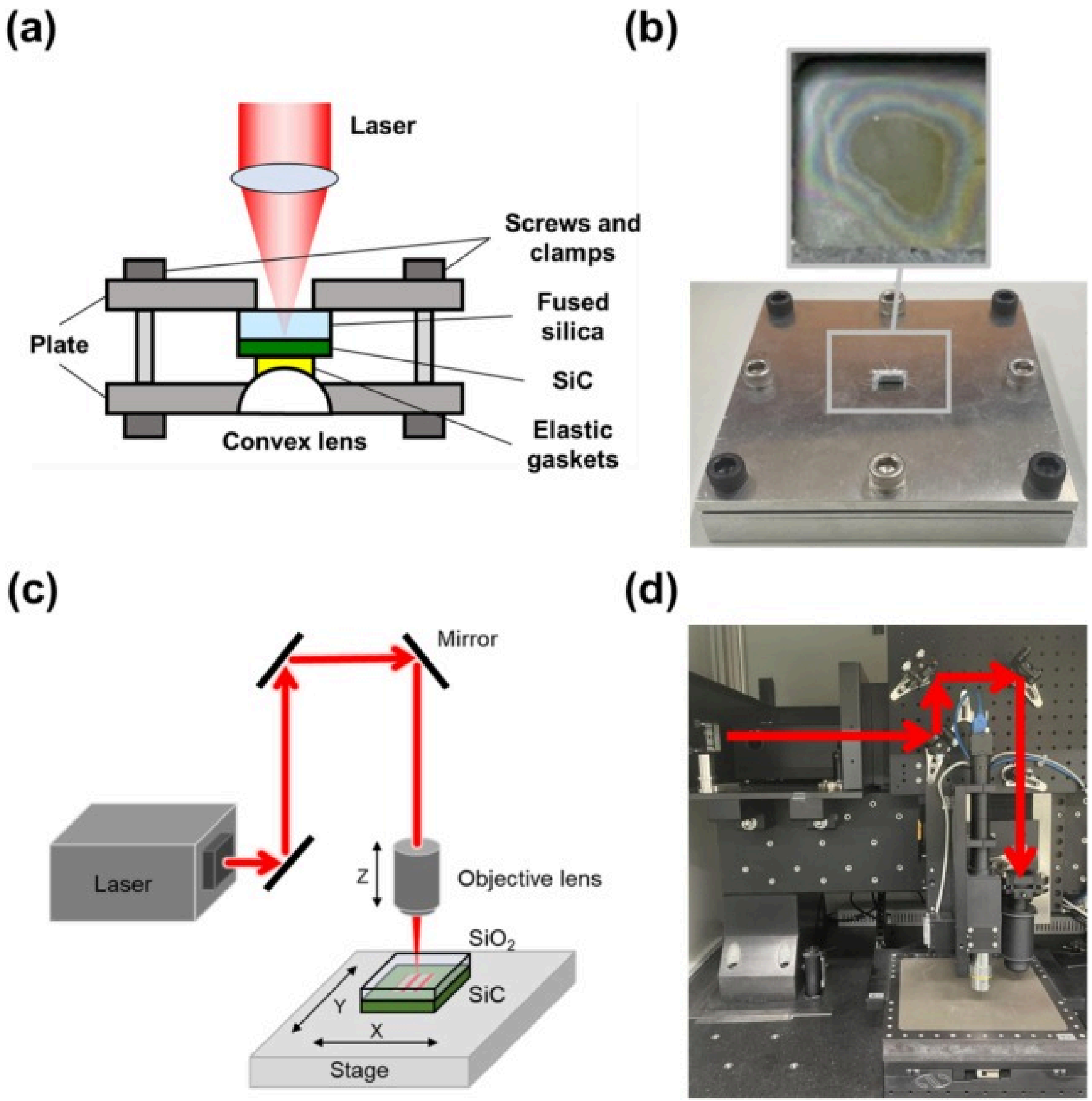

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Fig. 1. (a) Schematic and (b) picture of the fixture; (c) schematic and (d) picture of femtosecond laser system.

The experimental femtosecond laser system is illustrated in Fig. 1(c). The femtosecond fiber laser source (FS-MM, mRadian Inc.) had an output wavelength of 1030 nm, and a pulse width of approximately 221 fs with a Gaussian beam shape. The beam quality was M^2=1.3, with a maximum average power of 5 W and a repetition rate adjustable between 100 kHz and 1000 kHz. After being emitted, the laser beam was reflected by mirrors and focused onto the sample using a 50x objective lens with a numerical aperture (N.A.) of 0.65. The entrance aperture diameter of the objective was 6 mm, resulting in a focused spot size of 1.94 μm (at $1/e^2$), with a focal length of 14.5 mm and a depth of focus (double Rayleigh length) of 2.44 μm in air. Aerotech's A3200 software controlled the movement of the XY-axis stage and the Z-axis position of the objective lens for precise laser processing.

2.2. Research method

The primary experimental parameters in this study were the laser repetition rate and focal depth. These were adjusted while maintaining constant pulse energy and equivalent pulses across all experiments. As shown in equations (1), (2), laser power and scanning speed were directly proportional to the repetition rate. Therefore, as the repetition rate varied, both the laser power and scanning speed were adjusted proportionally to ensure consistent conditions.

$$J_0 = P_0/f_{rep} \tag{1}$$

$$N_{eff} = 2r_0 \times \frac{f_{rep}}{v} \tag{2}$$

where J_0 represents the pulse energy, P_0 is the laser power, f_{rep} is the repetition rate, N_{eff} represents the effective number of pulses, r_0 is the spot radius, and v is the scanning speed.

Previous research on femtosecond laser welding of SiC and fused silica has demonstrated that by controlling energy density and scanning distance, a shear strength of 15.1 MPa can be achieved [10]. That study used a pulse width of 240 fs, a laser wavelength of 800 nm, a repetition rate of 50 kHz, a focused spot size of 2.5 µm (at $1/e^2$), and a scanning speed of 0.5 mm/s. Under these conditions, the effective number of pulses was calculated to be 250.

In this study, we employed an infrared femtosecond laser and maintained the effective number of pulses at 250, consistent with the parameters reported in the previous study [10]. By adjusting the repetition rate and focal depth, lateral observations of the welding area were made to determine the optimal parameters. Table 1 outlines the different laser parameters corresponding to various repetition rates. Different repetition rates were associated with varying laser powers to maintain a constant energy density. The scanning speed was adjusted to fix the effective number of pulses at 250. The focal depth was defined as the surface of the fused silica (Z = 0) and was adjusted downward.

Table 1. Laser parameter design.

Repetition rate (kHz)	**100**	**300**	**1000**
Laser power (W)	0.1	0.3	1.0
Scanning speed (mm/s)	0.772	2.316	7.720

2.3. Measuring instruments

Both lateral and fractured surface observations of the welding area were conducted. The preparation process for lateral observations is shown in Fig. 2. The welding area was placed at the edge of the sample, embedded, ground, and then observed laterally. The samples were polished using 2000-grit sandpaper and a final polish with a 0.05 µm alumina suspension. For fractured surface observations, the samples were separated, and the surface morphology of the welding area was observed, as shown in Fig. 2.

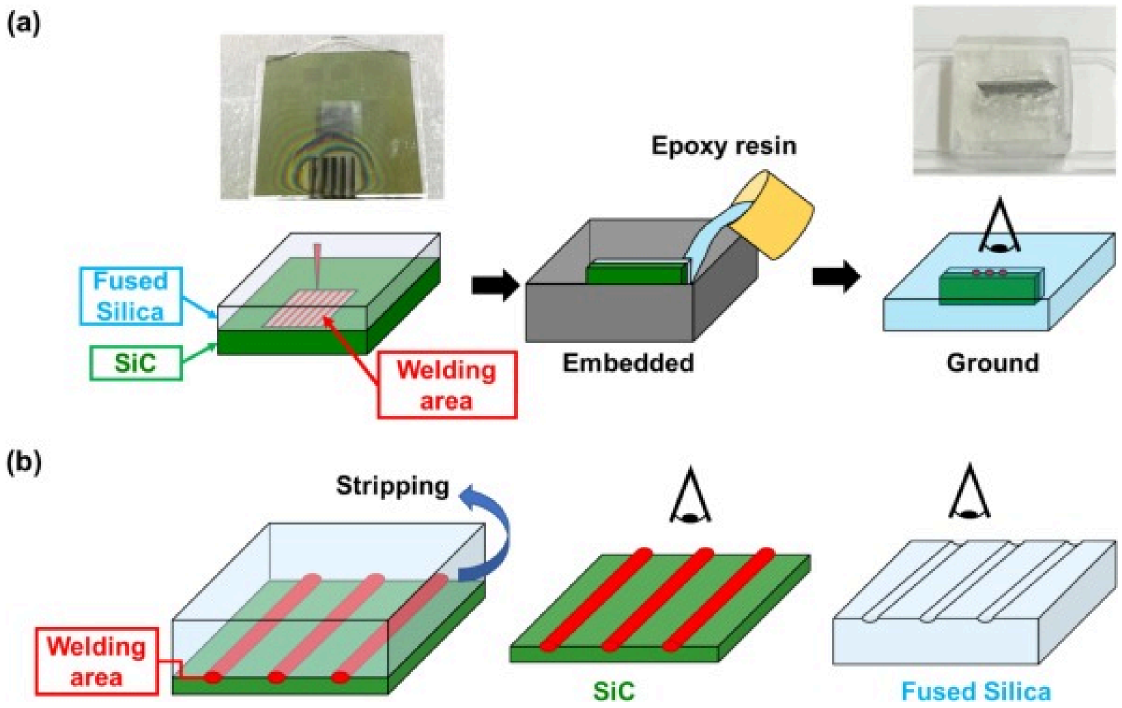

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Fig. 2. Schematic of (a) the lateral observation sample preparation process and (b) the fractured surface observation sample preparation process.

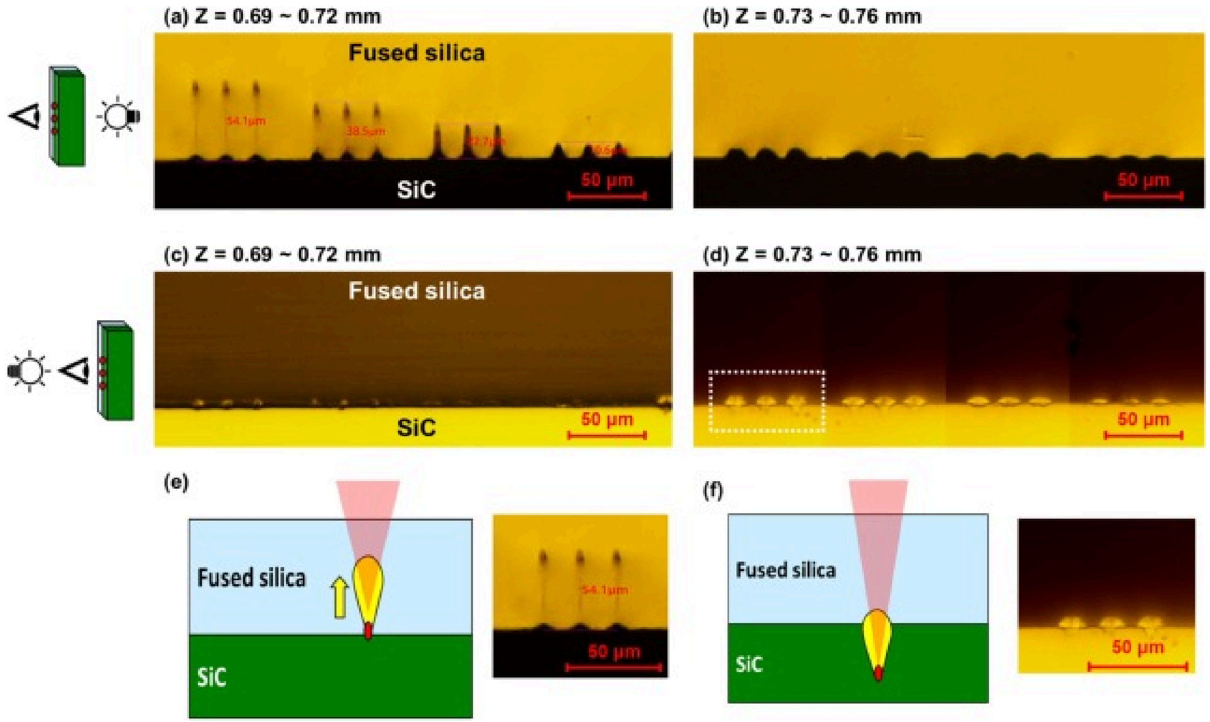

Fig. 3. (a)(b) Transmission mode and (c)(d) reflection mode OM images of the welding region with a repetition rate of 300kHz; (e)(f) schematic representation of the welding experiment and magnified OM images of (e) Z=0.69 and (f) Z=0.73mm.

Various instruments were employed for microstructure observation and elemental analysis. Optical microscopy (OM, Axioskop 40), scanning electron microscopy (SEM, SU-8010, Hitachi Inc.), and energy-dispersive X-ray spectroscopy (EDS, HORIBA X-max) were used to characterize the welding area.

3. Results and discussion

3.1. Line scanning: lateral observation

In this experiment, repetition rates of 100, 300, and 1000 kHz were used, and the focal depth was varied between 0.69 and 0.76 mm in increments of ΔZ = 0.01 mm. Three weld lines were produced under the same laser parameters. After welding, the samples were prepared using the lateral observation method (as shown in Fig. 2(a)). This study will first present the results obtained at a repetition rate of 300 kHz, followed by a comparison of experimental outcomes, as summarized in Fig. 4. Fig. 3 presents the OM images of the welded areas in both transmission and reflection modes at a repetition rate of 300 kHz. The transmission OM images were used to observe the laser focus position, while the reflection OM images revealed the characteristics of the welded areas.

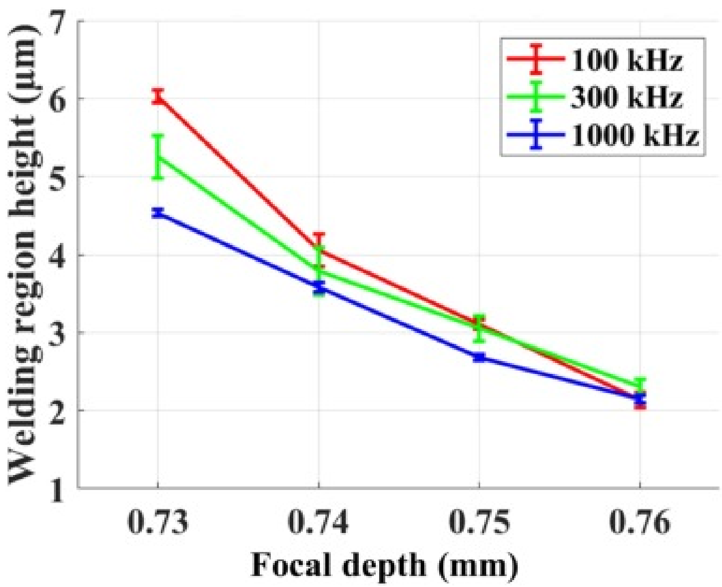

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Fig. 4. Measured height of the welding region at different focal depths and repetition rates (100 to 1000 kHz).

At a focal depth of 0.69 mm (Fig. 3(a)), a droplet-shaped modification region was observed in the fused silica. When the focal depth was increased from 0.73 to 0.76 mm (Fig. 3(b)), the top of the modification region approached the interface between the two materials. In contrast, Fig. 3(c) shows that no distinct welding region was formed at specific focal depths. However, at a focal depth of 0.73 mm (Fig. 3(d)), a welding region was observed at the interface (outlined by a white dashed box).

As shown in Fig. 3(a), when the focal depth (Z) is between 0.69 and 0.72 mm, the height of the modified region ranges from 54.1 µm to 10.6µm, which is significantly larger than the Rayleigh length in air, $R_{z,\ air}$ (1.22µm). In addition, the Rayleigh lengths within fused silica and SiC were calculated as approximate reference values: $R_{z,\ silica}$ (1.51 µm) and $R_{z,\ SiC}$ (2.52 µm) [11]. When the focal depth is set at the fused silica/SiC interface, the Rayleigh length can be reasonably assumed to lie between $R_{z,\ silica}$ and $R_{z,\ SiC}$.

These findings indicate that the modification phenomenon observed in Fig. 3(a) is not solely caused by avalanche ionization at the focal point. Instead, the experimental observations can be explained by the laser's nonlinear absorption, which initiates avalanche ionization at the focal point and generates an upwardly extending plasma column (Fig. 3(e)(f)) [7], [8]. In Fig. 3(e), this plasma column (marked by a yellow region) ultimately forms the droplet-shaped modification region seen in Fig. 3(a).

In Fig. 3(f), when the laser focus is positioned below the interface within the SiC material, avalanche ionization occurs at the focal point due to multiphoton absorption. The absorption front then propagates toward the incoming laser through the material interface, effectively preventing energy deposition beyond the focal depth [12]. This mechanism results in the formation of a droplet-shaped welding region. The high energy density at the upper part of the welding region causes the fused silica to melt and bond with dissociated SiC, thereby forming a welded area, as illustrated in Fig. 3(f).

In Fig. 3(d), transmission OM images reveal a distinct feature at the top of the modification region. The distance between the top of the welding region and the interface was measured to evaluate the welding region height. However, as shown in Fig. 3(a), for the focal depths between Z = 0.69 and 0.72 mm, the welding region remains entirely within the fused silica layer. In other words, the welding characteristics in this range do not occur at the interface. Therefore, this study does not analyze the welding region height for these focal depths.

Fig. 4 shows the heights of the welding regions measured at various focal depths (0.73 to 0.76 mm) and repetition rates (100, 300, and 1000 kHz). As shown in Fig. 4, the height of the welding region decreases as the focal depth increases. At Z = 0.76 mm, the height variation between different repetition rates was minimal, but at Z = 0.73 mm, the lower repetition rate (100 kHz) produced a taller welding region compared to the higher repetition rate (1000 kHz). Previous literature [13] has examined the influence of repetition rate on the position of the heat-affected zone (HAZ) in glass/glass welding. Under identical focusing conditions and pulse energy, changes in the repetition rate did not alter the HAZ position, which is similar to the observation at Z = 0.76 mm in this study. At Z = 0.73 mm, however, the observed height differences can be attributed to the optical Kerr effect [14], [15] or thermal lensing at higher repetition rates. These nonlinear effects may shift the laser focal point slightly downward at higher intensities, thereby reducing the welding region height.

The Kerr effect is commonly observed in nonlinear optical materials, where the refractive index n changes with laser intensity I, as expressed by the following equation:

$$n = n_0 + n_2 I \tag{3}$$

where n_0 is the linear refractive index, and n_2 is the second-order nonlinear refractive index.

In fused silica, the nonlinear refractive index n_2 is approximately 10^{-20} m^2/W [16], while in SiC it is around 10^{-19} m^2/W [17], [18]. This means that to induce significant changes in refractive index through the Kerr effect, the laser intensity must exceed 10^9 W /cm^2 for fused silica and 10^3 W /cm^2 for SiC. In this study, the highest laser intensity reached about

7×10^7 W/cm^2, insufficient to produce notable refractive index changes in fused silica but capable of affecting the refractive index in SiC, thereby influencing the focal point.

To further verify the mixing and diffusion of SiC and fused silica materials within the welding region, as shown in Fig. 5(a) to (d), SEM was used to capture images of the welding areas at various focal depths for the sample depicted in Fig. 3(d). It was observed that the welding region was predominantly located near the interface between the two materials. The height of the welding region, indicated by the white dashed line, decreased with increasing focal depth. An EDS line scan was performed along the yellow line through the welding region, moving from the SiC side toward the fused silica. The resulting elemental distributions are displayed in Fig. 5(e) to (h). Within the focal depth range of 0.74 to 0.76 mm, a peak in silicon (Si) concentration was identified. This peak transitioned from a stable concentration in the SiC region to a lower concentration, mixed with oxygen (O), in the SiO_2 region. The peak distribution (highlighted by the black circle in the figure) shifted towards the SiC side as the focal depth increased.

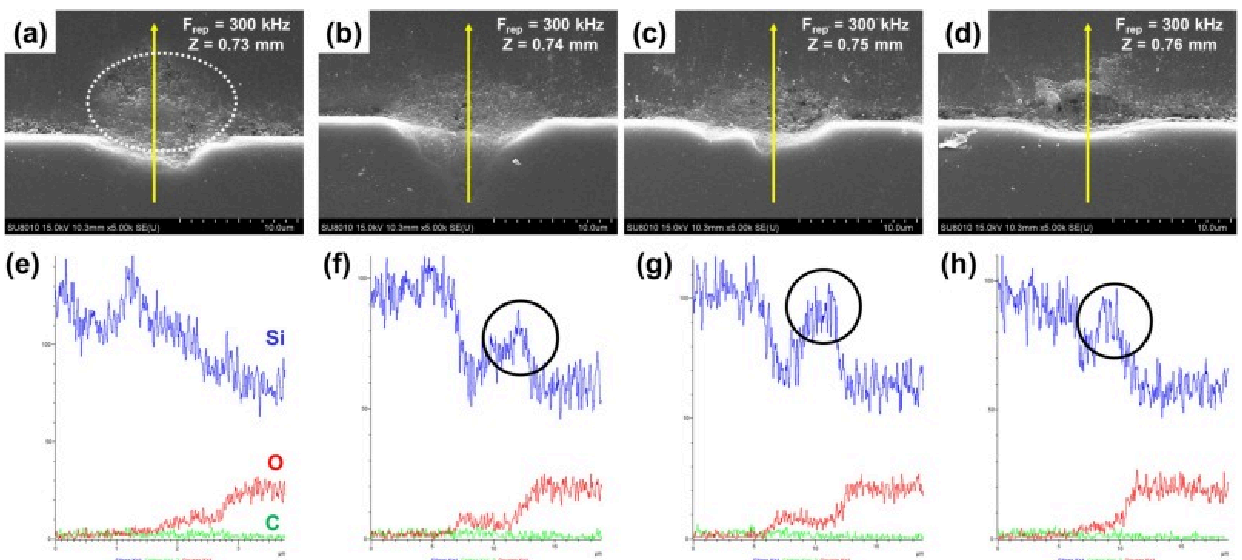

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Fig. 5. SEM images of the welding region (Z = 0.73 ~ 0.76 mm) and corresponding EDS line scan results. (a)-(d) SEM images, (e)-(h) EDS line scan results, where the blue line represents Si, the red line represents O, and the green line represents C.

To further understand the EDS line scan results, EDS surface scans were performed on samples with prominent Si peak distributions, as shown in Fig. 6. Using Z=0.74 mm as an example, Fig. 6(b) demonstrates that Si was predominantly distributed around the outer regions of the welding area, with higher concentrations along both sides, while the central region showed a lower concentration of Si. Fig. 6(c) demonstrates that oxygen (O) was uniformly distributed throughout the welding region, while Fig. 6(d) indicates that carbon (C) was concentrated in the center of the welding area. The EDS surface scan results for Si at different focal depths are displayed in Fig. 7. As the focal depth increased, the height of the welding region decreased, and the distinct elemental distribution patterns became less pronounced.

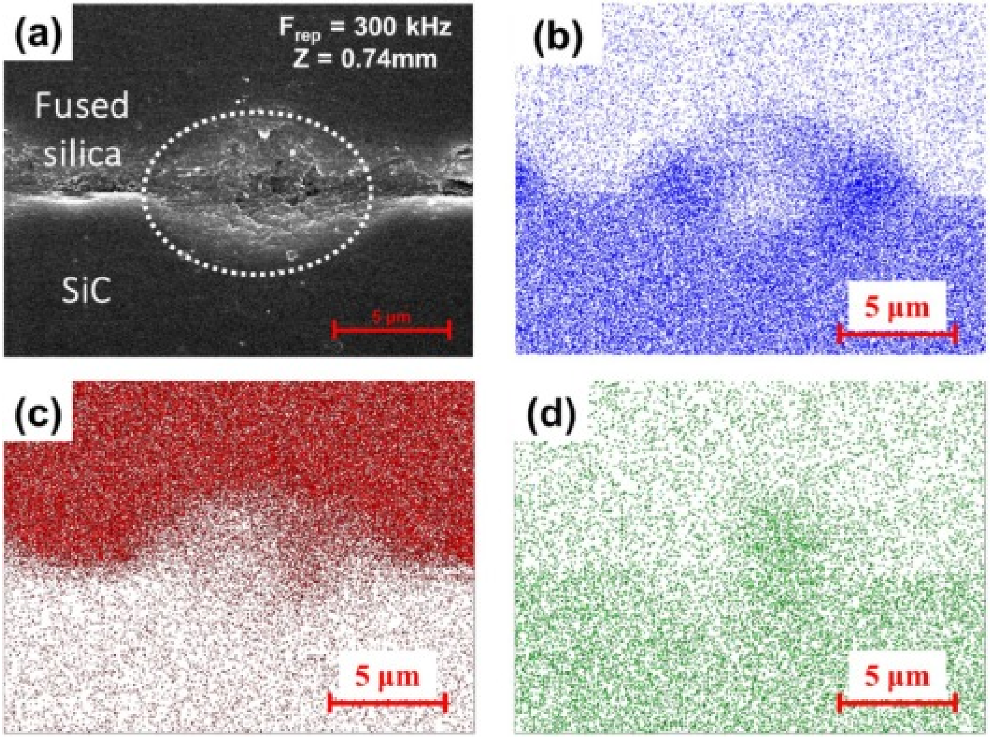

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Fig. 6. EDS results of the welding region (Z = 0.74mm): (a) SEM image, (b) Si distribution, (c) O distribution, (d) C distribution.

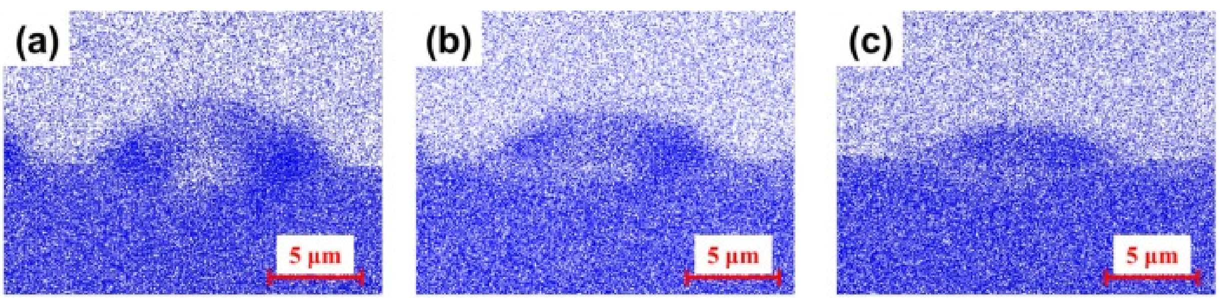

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Fig. 7. Si element EDS results at different focal depths: (a) 0.74 mm, (b) 0.75 mm, and (c) 0.76 mm.

It is hypothesized that under laser irradiation, the SiC within the welding region decomposes into liquid silicon (Si) and carbon (C). According to the literature [19], SiC can decompose into silicon and carbon at approximately 2840 K, and the temperatures above 2840 K can be readily achieved even with a single femtosecond laser pulse [20], [21]. Meanwhile, fused silica (SiO_2) either melts or vaporizes, producing liquid silicon (Si) and oxygen (O_2). When liquid Si is subjected to plasma and isostatic conditions, it mixes with liquid SiO_2 [22]. The migration rates of the ionic elements are influenced by activation energy [9]. Since the migration rate of Si is faster than that of C, Si tends to accumulate at the outer edges of the welding area, while C concentrates in the center. As the welding region shrinks with changes in focal depth, the migration distances of the elements decrease, making it more difficult to observe clear distinctions in elemental distribution.

3.2. Line scanning: fractured surface observation

Fig. 8 shows the SEM images of the SiC and fused silica surfaces after the samples were manually separated by stripping, with the images taken at a 15° tilt. In Fig. 8(a), the presence of welding material adhering to the surface of the SiC sample can be seen, while in Fig. 8(b), a groove structure is observed on the surface of the fused silica. Additionally, when observing the welding track from the SiC side, as shown in Fig. 8(a), the welding line appears raised on both sides and concave in the middle. This observation aligns with the earlier EDS surface scan results (Fig. 6(b)), where a higher concentration of Si elements was detected on both sides of the welding region.

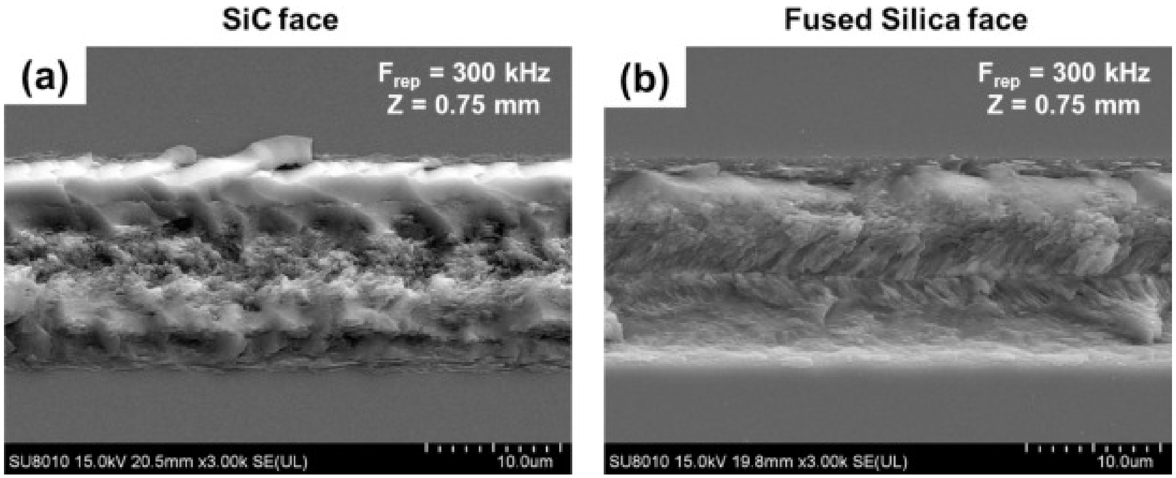

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Fig. 8. SEM images of the separated samples, taken at a 15° tilt: (a) SiC surface and (b) fused silica surface.

To verify the hypothesis concerning the formation of the welding track shape, welding tracks produced at different focal depths (0.75 mm and 0.725 mm) were analyzed. The tracks were divided into three distinct regions on the SiC surface: the outer area of the track, the edge of the track, and the center of the track. Fig. 9 shows SEM observations of these regions. At a focal depth of 0.75 mm (Fig. 9(a)), the welding track appears raised on both sides and concave in the middle. In contrast, the welding track exhibits a raised profile in the middle at a focal depth of 0.725 mm (Fig. 9(b)).

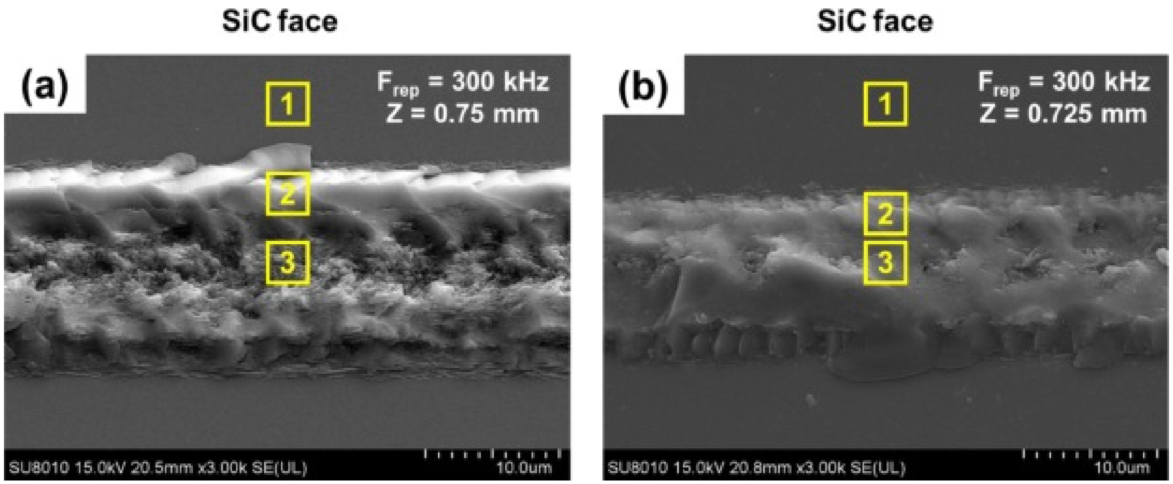

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Fig. 9. SEM images of the SiC surface at different focal depths: (a) 0.75 mm and (b) 0.725 mm.

Table 2 presents the EDS elemental analysis of each region shown in Fig. 9. Both focal depths exhibit similar elemental compositions in Region 1 (primarily composed of C and Si), representing the original SiC material, and in Region 3 (mainly composed of O and Si), which corresponds to the welding material adhering to the SiC surface, primarily originating from the fused silica. In Region 2, however, the Si content at a focal depth of 0.75 mm is 82.42 wt%, significantly higher than the 54.51 wt% observed at a focal depth of 0.725 mm. This result confirms the correlation between the separation profile of the welding region and the Si element distribution.

Table 2. EDS elemental analysis for different regions.

Z = 0.75 mm				**Z = 0.725 mm**			
Region 1	Element	Weight%	Atomic%	**Region 1**	Element	Weight%	Atomic%

Z = 0.75 mm				**Z = 0.725 mm**			
	C	45.95	66.53		C	52.72	72.28
	Si	54.05	33.47		Si	47.28	27.72
Region 2	O	17.58	27.24	**Region 2**	O	45.49	59.44
	Si	82.42	72.76		Si	54.51	40.56
Region 3	O	30.00	45.64	**Region 3**	O	40.77	54.72
	Si	61.49	53.29		Si	59.23	45.28

3.3. Area scanning

In this section, fully welded samples were prepared, with a scan area of 3×3 mm^2 and a scanning interval of 0.02 mm. After applying a stripping force to the welded samples (Z = 0.75 mm, 300 kHz), it was observed that fractures initiated in the fused silica region rather than in the welding area, as shown in Fig. 10(a). Notably, the fractured fused silica remained adhered to the surface of the SiC sample, forming island-like structures, as illustrated in Fig. 10(b).

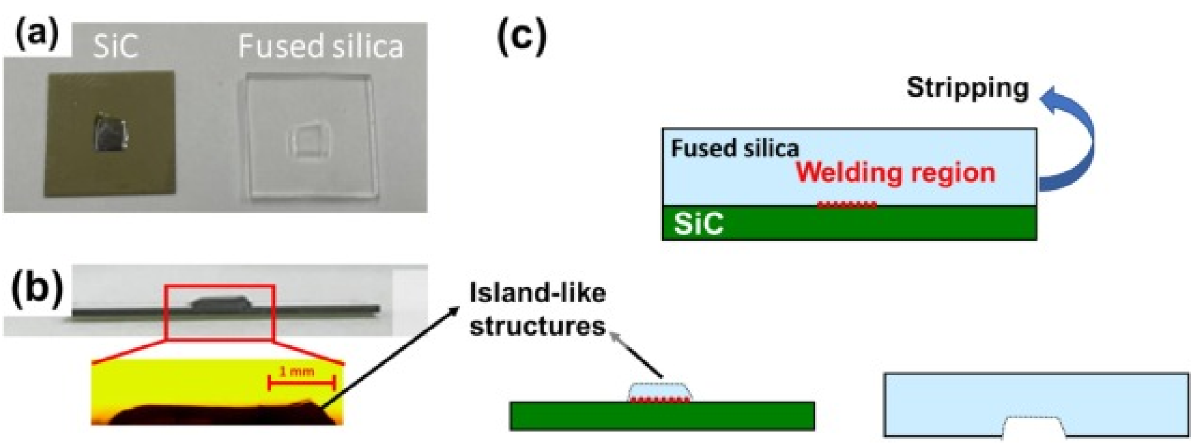

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Fig. 10. Fracture of the samples after manual separation: (a) top view, (b) side view, and (c) schematic representation.

To investigate the issue of fused silica fracturing outside the welding region, the parameters were adjusted, and the changes in the thickness of the island-like structures were measured before and after the adjustments. The objective was to identify optimal parameters that would minimize the extent of fracturing around the welding area. The adjustments to the laser parameters were made in several key areas: (1) varying the focal depth within the previously demonstrated range capable of forming effective welds, as indicated in Fig. 3(d)); (2) increasing the repetition rate to enhance heat accumulation effects [23]; (3) increasing the scanning interval to reduce overlap between adjacent heat-affected zones; and (4) lowering the laser power to minimize the heat-affected zone in the welding area.

The results of these parameter adjustments on the thickness of the island-like structures are summarized in Table 3. Variations in focal depth, repetition rate, and scanning interval caused changes in the thickness of the island-like structures by less than 12%. However, adjusting the laser power produced a much more significant effect, with changes exceeding 30% in thickness. In subsequent experiments, the laser power was reduced to 0.15W, but no welding occurred, as the energy was below the threshold required for effective welding, and no visible machining marks were left on the contact surface.

Table 3. The effect of parameter adjustments on the thickness of island-like structures.

Adjust parameters	**Parameter value**	**Island structure thickness (μm)**	**Island structure thickness ratio changes (%)**
Focal depth (mm)	0.75→0.725	261.4	+4.56

Adjust parameters	Parameter value	Island structure thickness (μm)	Island structure thickness ratio changes (%)
Repetition rate (kHz)	300→1000	279.5	+11.80
Scanning interval (μm)	20→40	271.2	+8.48
Laser power (W)	0.3→0.2	173.4	−30.64

However, the welded specimens still tended to separate around the fused silica melt pool, thus no further tensile strength tests were performed. It is hypothesized that differences in thermal expansion and contraction between the welded and non-welded regions of the fused silica induced stress differentials, which led to the formation of microcracks and ultimately caused the material to fracture.

Moreover, previous studies [24], [25] on femtosecond laser welding of fused silica to fused silica have reported disruptions (hollow cavities) within the welding region and smaller dark modifications in the molten zone. These disruptions are located near the focusing lens, where the temperature is highest and solidification occurs last. The smaller dark modifications are likely gas bubbles formed during laser irradiation, subsequently rising within the molten pool due to buoyancy forces.

4. Conclusion

This study employed an infrared femtosecond laser to weld SiC and fused silica (SiO_2), adjusting the laser repetition rate and focal depth to obtain several key findings. A well-defined welding region was established at the interface between the SiC and fused silica at the appropriate focal depth. Silicon (Si) elements accumulated primarily along the edges of the welding region, especially on both sides, while carbon (C) concentrated in the center. The welding region separated from the fused silica and adhered to the SiC, with its morphology influenced by the distribution of elements. At a focal depth of 0.75 mm, for instance, the welding region tended to split along the contours of peak Si element distribution. Fractures commonly occurred in the fused silica outside the welding area, forming island-like structures around 250μm thick. The thickness of these island-like structures was effectively reduced by about 30% by lowering the laser power.

Further investigations, including experiments at higher repetition rates (> 1 MHz) [24] or altering the direction of the laser incidence (currently from fused silica toward SiC), will be necessary to better understand these processes and clarify why the welded specimens continued to separate around the fused silica melt pool.

CRediT authorship contribution statement

Yu-Guo Jiang: Writing – original draft, Data curation, Conceptualization. Jia-Fan Kuo: Data curation, Conceptualization. **Chung-Wei Cheng:** Writing – original draft, Funding acquisition, Conceptualization. **An-Chen Lee:** Supervision, Funding acquisition. **Yasuhiro Okamoto:** Writing – review & editing, Supervision, Investigation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Chung-Wei Cheng reports financial support was provided by National Science and Technology Council of Republic of China.

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Data availability

No data was used for the research described in the article.

References

[1] I. Kazuyoshi, T. Takayuki
Ultrafast laser microwelding for transparent and heterogeneous materials
Proc.spie (2008), Article 68810V
Google Scholar ↗

[2] K. Sugioka, Y. Cheng
Femtosecond laser processing for optofluidic fabrication
Lab Chip, 12 (19) (2012), pp. 3576-3589
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[3] T. Tamaki, W. Watanabe, J. Nishii, K. Itoh
Welding of transparent materials using femtosecond laser pulses
Jpn. J. Appl. Phys., 44 (5L) (2005), p. L687
View in Scopus ↗ Google Scholar ↗

[4] T. Tamaki, W. Watanabe, K. Itoh
Laser micro-welding of transparent materials by a localized heat accumulation effect using a femtosecond fiber laser at 1558 nm
Opt. Express, 14 (22) (2006), pp. 10460-10468
View in Scopus ↗ Google Scholar ↗

[5] I. Miyamoto, A. Horn, J. Gottmann, D. Wortmann, F. Yoshino
Fusion welding of glass using femtosecond laser pulses with high-repetition rates
J. Laser Micro / Nanoeng., 2 (2007), p. 57
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[6] I. Miyamoto, K. Cvecek, M. Schmidt
Crack-free conditions in welding of glass by ultrashort laser pulse
Opt. Express, 21 (2013), pp. 14291-14302

View in Scopus ↗ Google Scholar ↗

[7] I. Miyamoto, K. Cvecek, M. Schmidt
Evaluation of nonlinear absorptivity in internal modification of bulk glass by ultrashort laser pulses
Opt. Express, 19 (2011), pp. 10714-10727
View in Scopus ↗ Google Scholar ↗

[8] I. Miyamoto, K. Cvecek, Y. Okamoto, M. Schmidt
Internal modification of glass by ultrashort laser pulse and its application to microwelding
Appl. Phys. A, 114 (2013)
Google Scholar ↗

[9] S. Huang, R. Chen, H. Zhang, J. Ye, X. Yang, J. Sheng
A study of welding process in connecting borosilicate glass by picosecond laser pulses based on response surface methodology
Opt. Laser Technol., 131 (2020), Article 106427
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[10] G. Zhang, J. Bai, W. Zhao, K. Zhou, G. Cheng
Interface modification based ultrashort laser microwelding between SiC and fused silica
Opt. Express, 25 (3) (2017), pp. 1702-1709
View in Scopus ↗ Google Scholar ↗

[11] L. Ionel, F. Jipa, A. Bran, E. Axente, G. Popescu-Pelin, F. Sima, K. Sugioka
Effect of varied beam diameter of picosecond laser on Foturan glass volume microprocessing
Opt. Express, 32 (11) (2024), pp. 20109-20118
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[12] C. Carr, J. Bude, P. DeMange
Laser-supported solid-state absorption fronts in silica
Phys. Rev. B—Condensed Matter and Mater. Phys., 82 (18) (2010), Article 184304
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[13] P.K. Sahoo, T. Feng, J. Qiao
Dynamic pulse propagation modelling for predictive femtosecond-laser-microbonding of transparent materials
Opt. Express, 28 (21) (2020), pp. 31103-31118
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[14] M. Chambonneau, Q. Li, V.Y. Fedorov, M. Blothe, K. Schaarschmidt, M. Lorenz, S. Tzortzakis, S. Nolte
Taming ultrafast laser filaments for optimized semiconductor–metal welding
Laser Photonics Rev., 15 (2) (2021), Article 2000433
View in Scopus ↗ Google Scholar ↗

[15] M. Chambonneau, Q. Li, M. Blothe, S.V. Arumugam, S. Nolte
Ultrafast laser welding of silicon
Adv. Photonics Res., 4 (5) (2023), Article 2200300
View in Scopus ↗ Google Scholar ↗

[16] T. Olivier, F. Billard, H. Akhouayri
Nanosecond Z-scan measurements of the nonlinear refractive index of fused silica
Opt. Express, 12 (7) (2004), pp. 1377-1382
View in Scopus ↗ Google Scholar ↗

[17] M. Sheik-Bahae, D.J. Hagan, E.W. Van Stryland
Dispersion and band-gap scaling of the electronic Kerr effect in solids associated with two-photon absorption

Phys. Rev. Lett., 65 (1) (1990), p. 96

View in Scopus ↗ Google Scholar ↗

[18] X. Guo, Z. Peng, P. Ding, L. Li, X. Chen, H. Wei, Z. Tong, L. Guo
Nonlinear optical properties of 6H-SiC and 4H-SiC in an extensive spectral range
Optical Materials Express, 11 (4) (2021), pp. 1080-1092

Crossref ↗ View in Scopus ↗ Google Scholar ↗

[19] K. Daviau, K.K. Lee
Decomposition of silicon carbide at high pressures and temperatures
Phys. Rev. B, 96 (17) (2017), Article 174102

Crossref ↗ View in Scopus ↗ Google Scholar ↗

[20] G. Tsibidis, L. Mouchliadis, M. Pedio, E. Stratakis
Modeling ultrafast out-of-equilibrium carrier dynamics and relaxation processes upon irradiation of hexagonal silicon carbide with femtosecond laser pulses
Phys. Rev. B, 101 (7) (2020)

Google Scholar ↗

[21] Y.-H. Liu, C.-W. Cheng
4H-SiC surface modified by a green femtosecond laser at low fluence
J. Laser Micro Nanoeng., 18 (3) (2023), pp. 157-160

Google Scholar ↗

[22] J. Zeng, Q. Zhang, J. Zhao, Y. Cai, C. Chu, Y. Zhu, J. Xu
The formed surface characteristics of SiCf/SiC composite in the nanosecond pulsed laser ablation
Ceram. Int., 48 (24) (2022), pp. 36860-36870

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[23] S.M. Eaton, H. Zhang, P.R. Herman, F. Yoshino, L. Shah, J. Bovatsek, A.Y. Arai
Heat accumulation effects in femtosecond laser-written waveguides with variable repetition rate
Opt. Express, 13 (12) (2005), pp. 4708-4716
View in Scopus ↗ Google Scholar ↗

[24] S. Richter, F. Zimmermann, A. Tünnermann, S. Nolte
Laser welding of glasses at high repetition rates–Fundamentals and prospects
Opt. Laser Technol., 83 (2016), pp. 59-66
View PDF View article View in Scopus ↗ Google Scholar ↗

[25] S. Richter, S. Döring, F. Burmeister, F. Zimmermann, A. Tünnermann, S. Nolte
Formation of periodic disruptions induced by heat accumulation of femtosecond laser pulses
Opt. Express, 21 (13) (2013), pp. 15452-15463
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